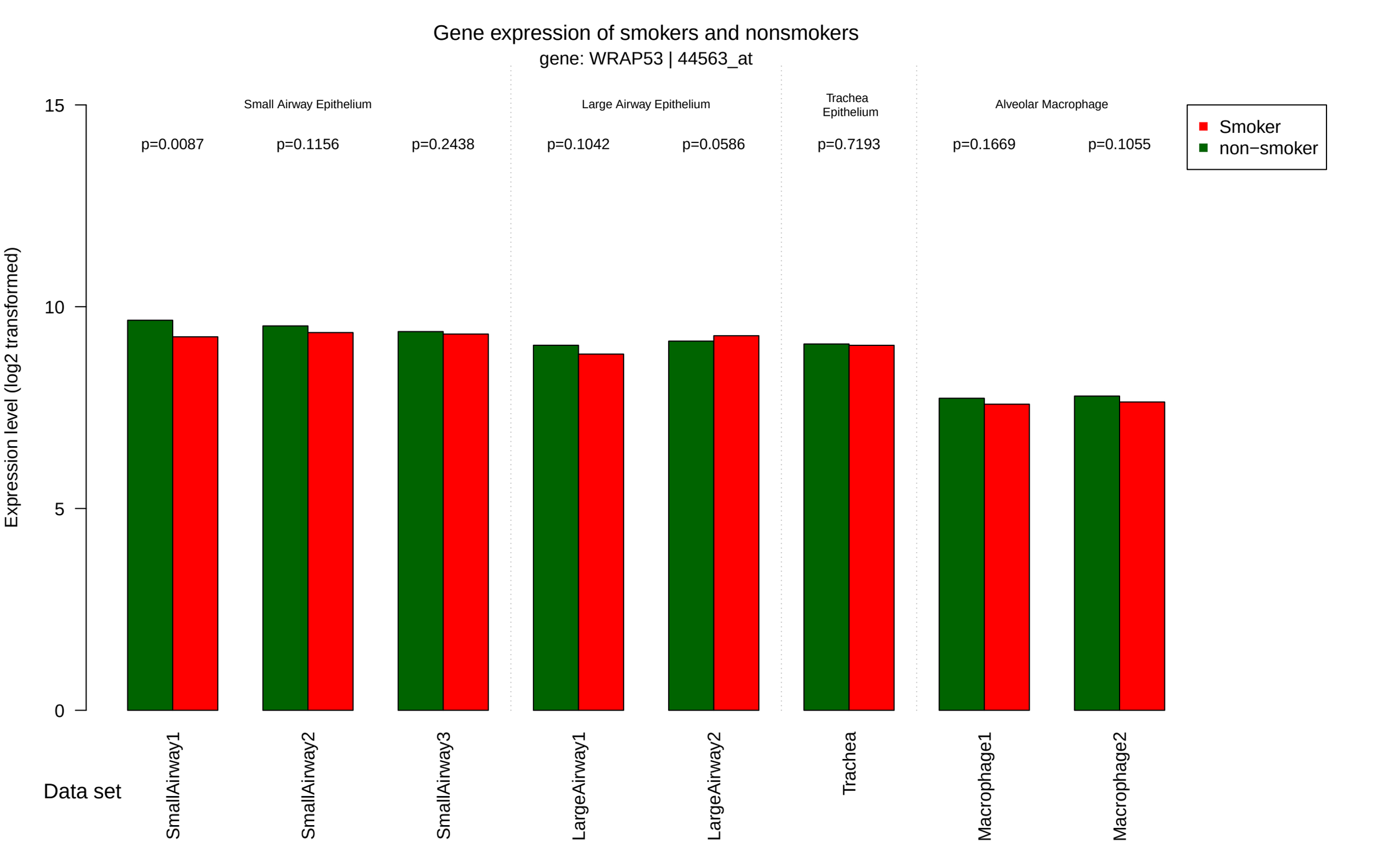
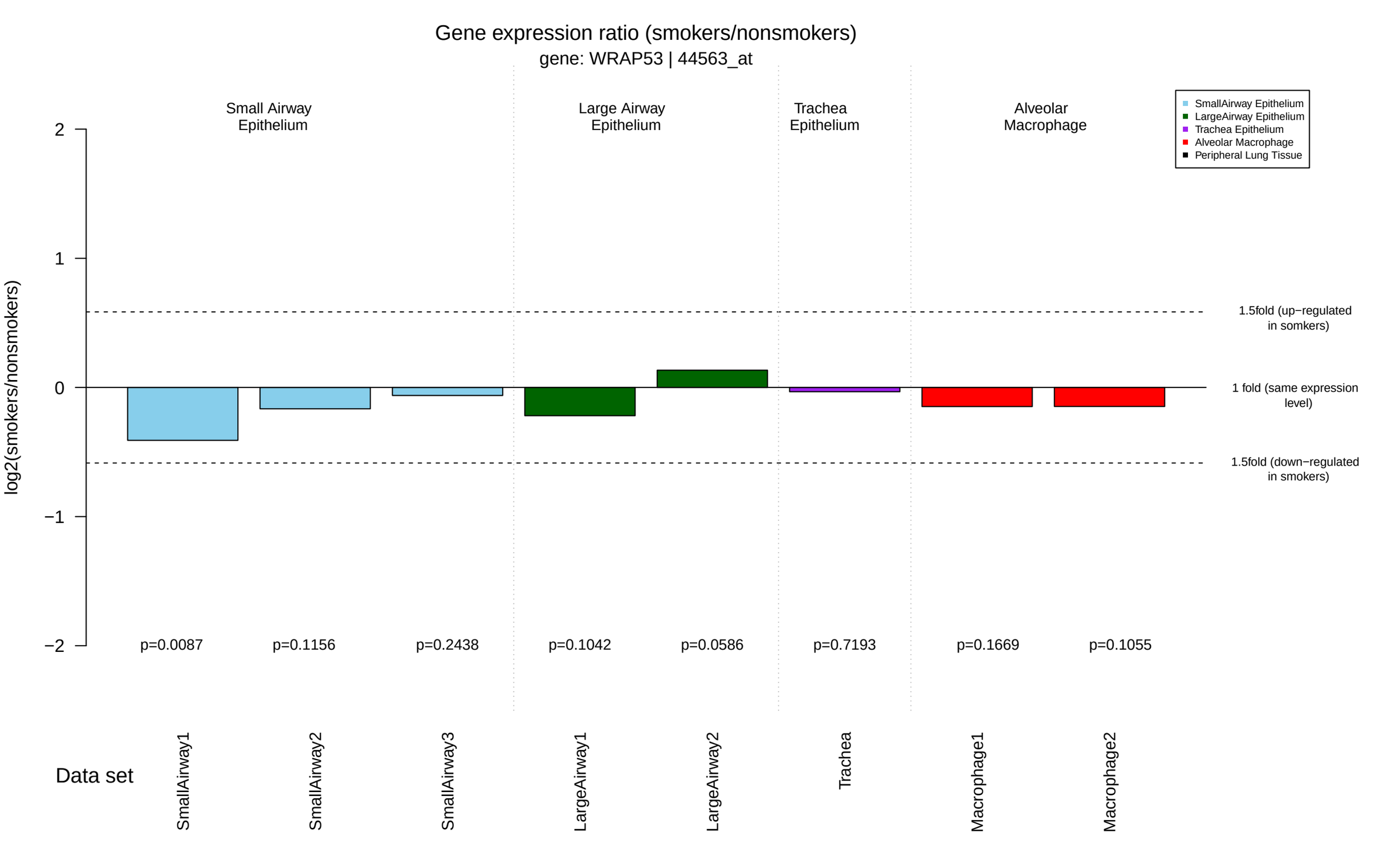
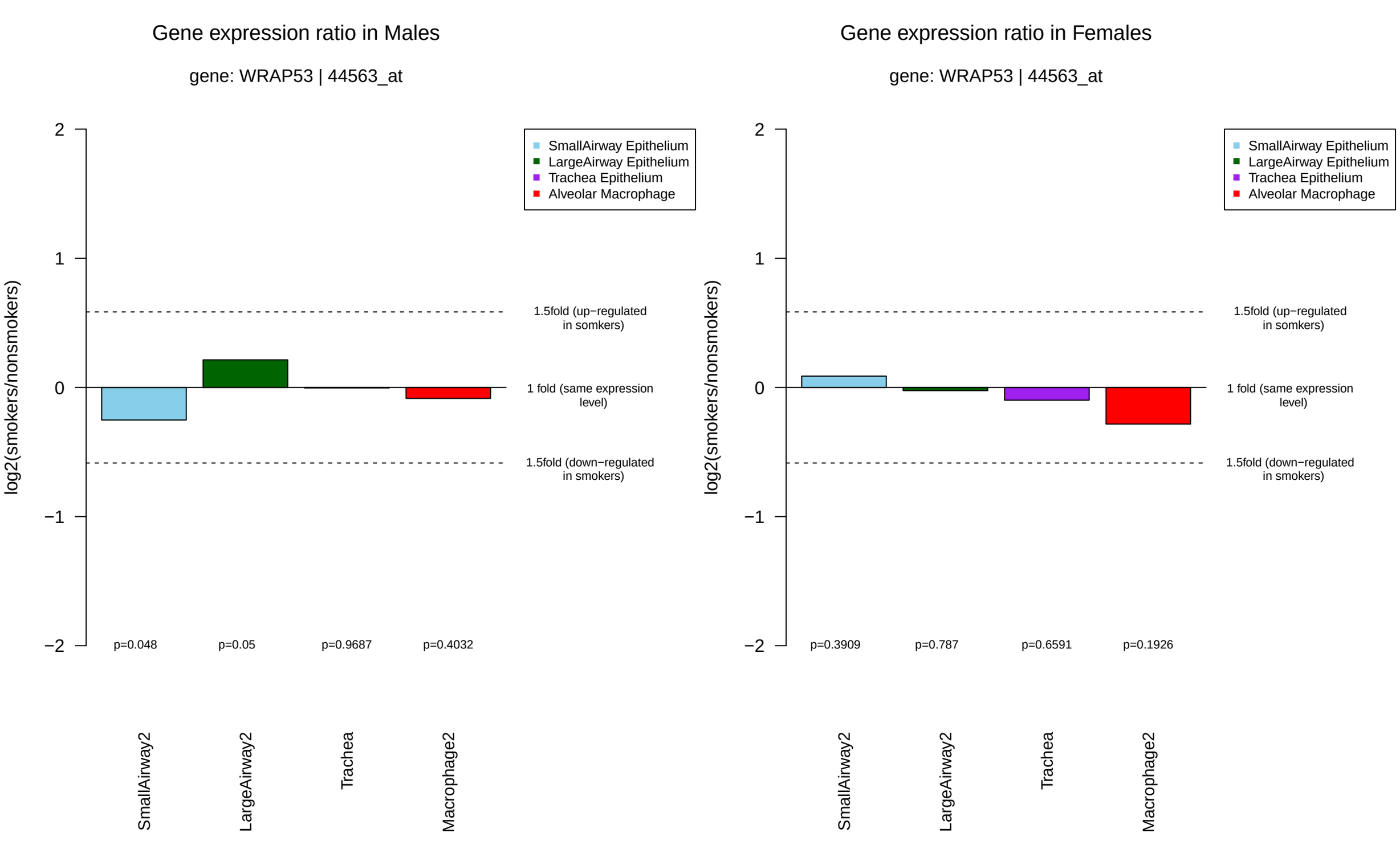
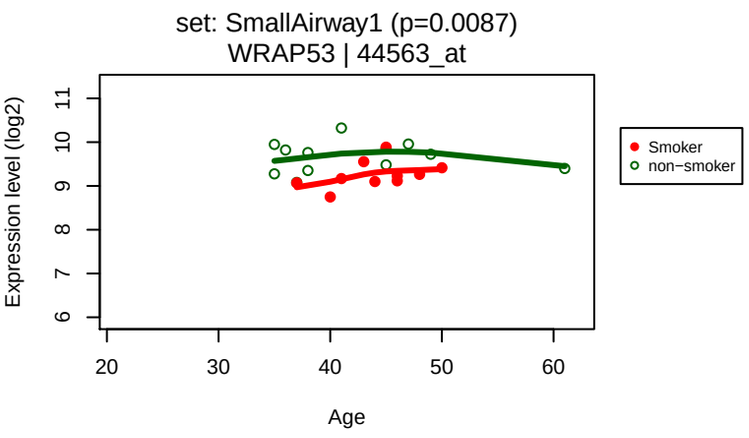
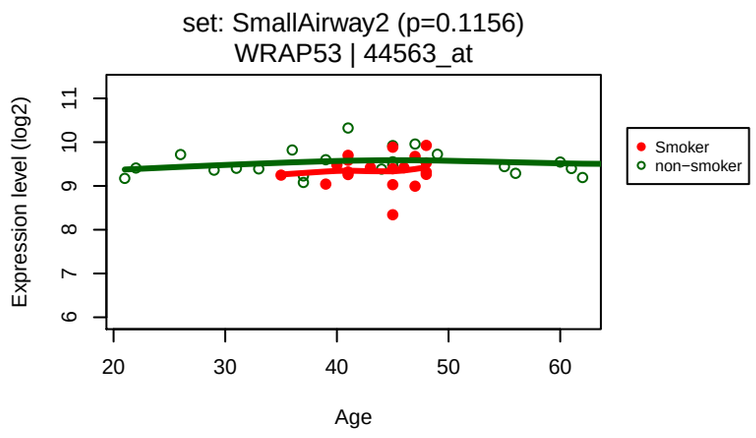
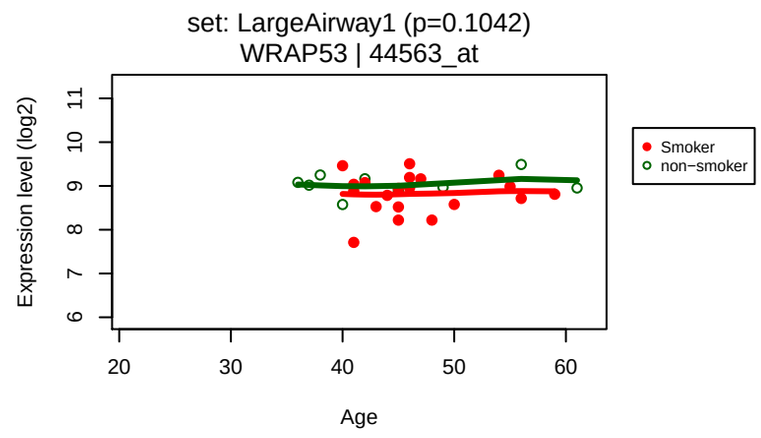

Gene expression of smokers and nonsmokers
gene: WRAP53 | 44563_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0087
p=0.1156
p=0.2438
p=0.1042
p=0.0586
p=0.7193
p=0.1669
p=0.1055
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

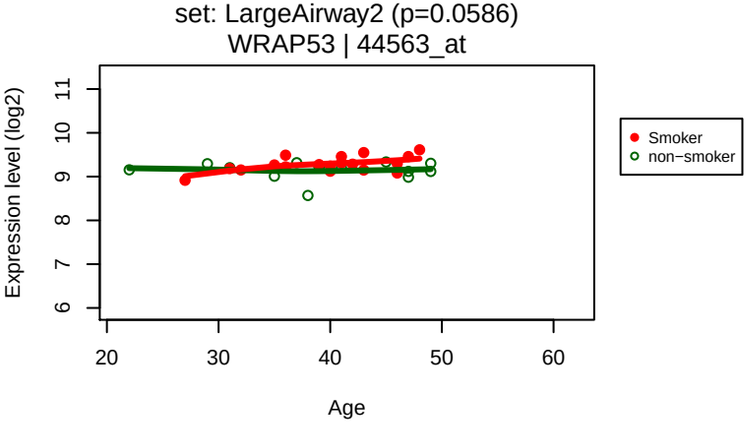

Gene expression ratio (smokers/nonsmokers)
gene: WRAP53 | 44563_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0087
p=0.1156
p=0.2438
p=0.1042
p=0.0586
p=0.7193
p=0.1669
p=0.1055
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

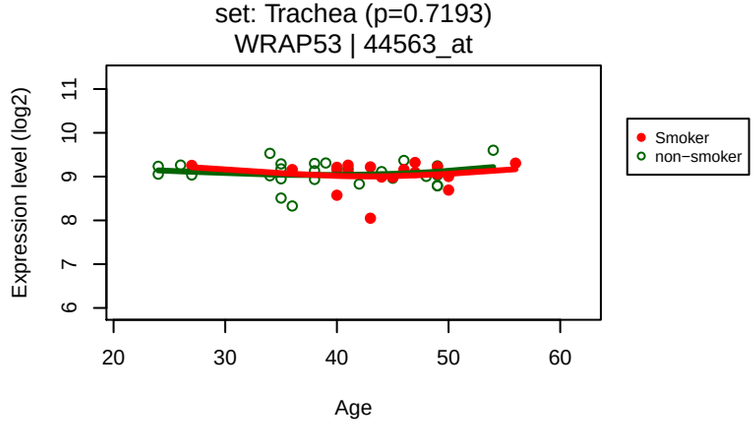

Gene expression ratio in Males
gene: WRAP53 | 44563_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.048
p=0.05
p=0.9687
p=0.4032
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WRAP53 | 44563_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3909
p=0.787
p=0.6591
p=0.1926
SmallAirway2
LargeAirway2
Trachea
Macrophage2

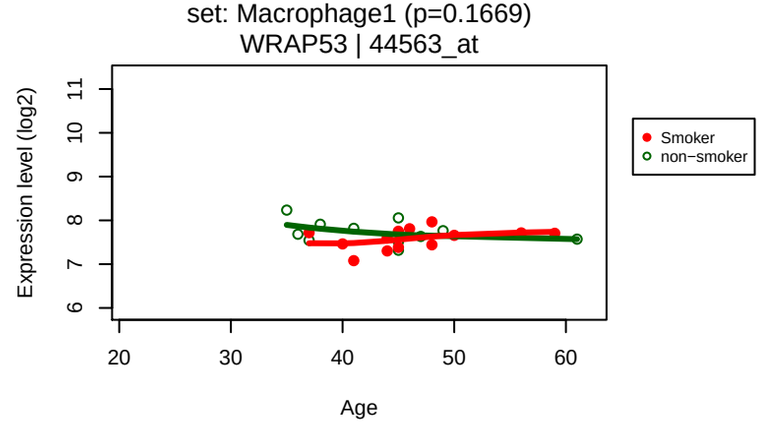

set: SmallAirway1 (p=0.0087)
WRAP53 | 44563_at
Expression level (log2)
Age
Smoker
non−smoker

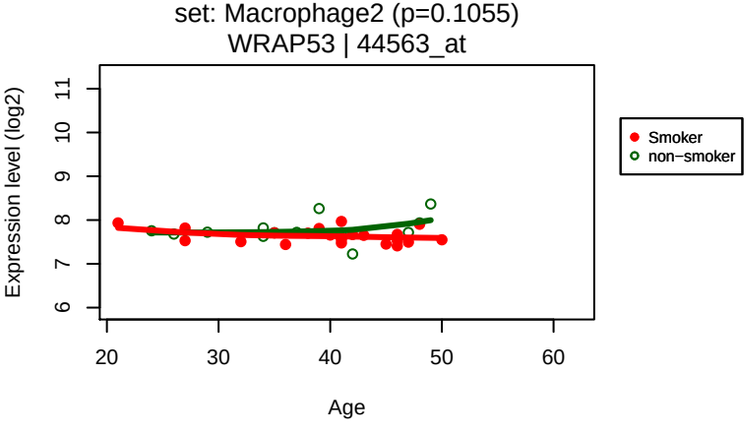

set: SmallAirway2 (p=0.1156)
WRAP53 | 44563_at
Expression level (log2)
Age
Smoker
non−smoker

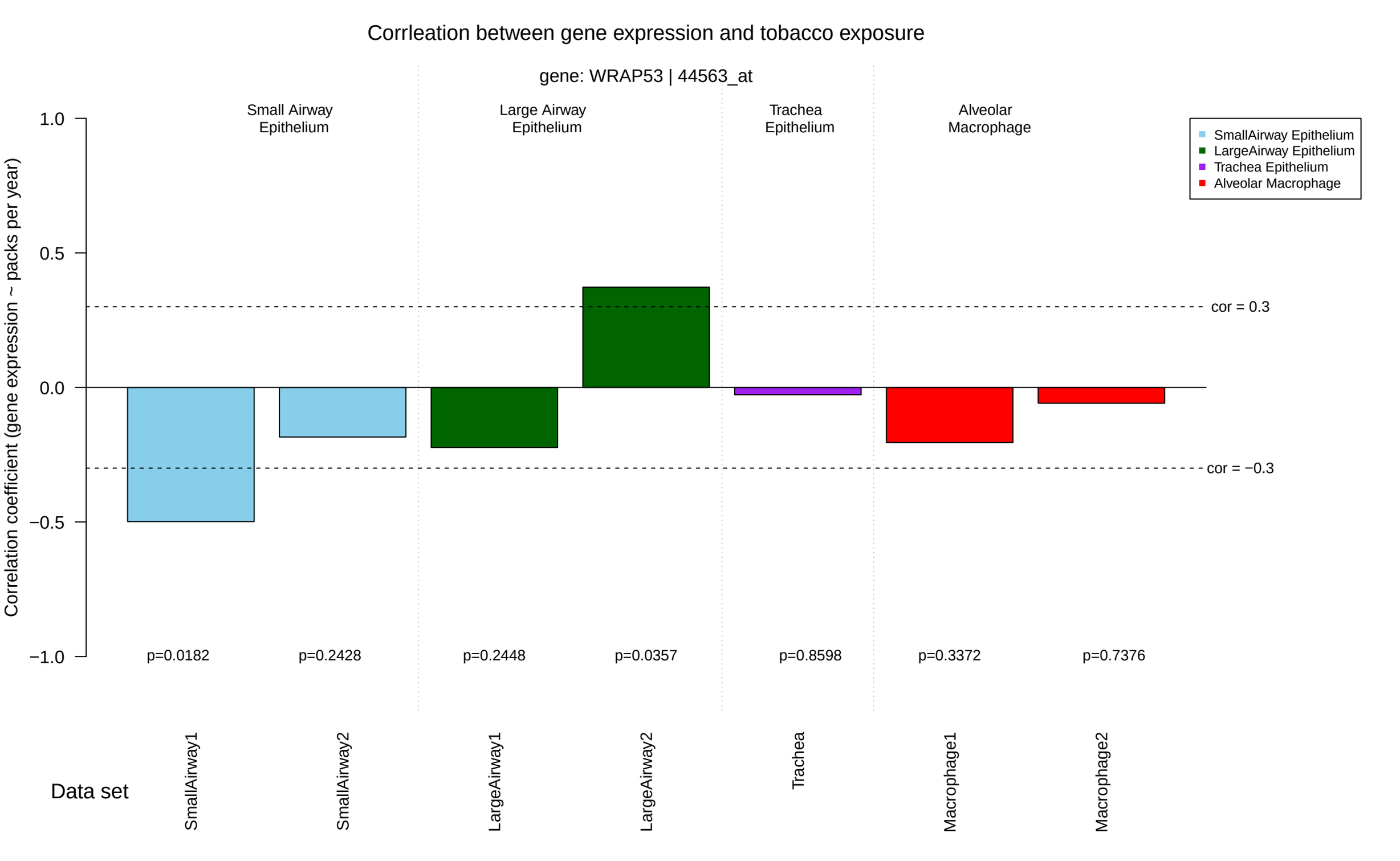

set: LargeAirway1 (p=0.1042)
WRAP53 | 44563_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0586)
WRAP53 | 44563_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7193)
WRAP53 | 44563_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1669)
WRAP53 | 44563_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1055)
WRAP53 | 44563_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WRAP53 | 44563_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0182
p=0.2428
p=0.2448
p=0.0357
p=0.8598
p=0.3372
p=0.7376
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2